

Written Report

DA-MATH 220 Final Project, Spring 2025

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Introduction

IMO and PISA Background

The International Mathematical Olympiad (IMO) stands as the world's most prestigious mathematics competition for high school students, bringing together elite young problem-solvers from over 100 countries in an annual display of mathematical brilliance (Yeo, 2006). Participants compete individually on a two-day exam featuring six deeply challenging problems covering topics such as algebra, number theory, geometry, and combinatorics. The competition not only showcases mathematical talent but also reflects broader educational systems' ability to identify, train, and support high-achieving students.

The first IMO competition was held in Romania and consisted of countries within the Soviet bloc such as Poland, Bulgaria, Hungary, Romania, Czechoslovakia, and a few others. It slowly began to incorporate more countries outside of Eastern Europe and into other continents until it reached the around 100 participating countries today. The composition of the teams has changed from up to 8 members to the 6 per country that are typically chosen from within national competitions within each own country. The competition has grown in popularity and has become a place for bright young minds to represent their country and themselves in a global stage of learning.

Alongside the IMO, the **Programme for International Student Assessment (PISA)**, coordinated by the OECD, serves as a global benchmark for evaluating the academic performance of 15-year-olds in mathematics, science, and reading literacy (OECD). Unlike the Olympiad, which targets top-performing students, PISA offers a snapshot of a country's overall educational effectiveness. Together, these two assessments provide a unique opportunity to explore the relationship between general educational outcomes and high-end academic performance.

This project investigates the connections between a country's overall PISA results and its performance at the IMO. Specifically, we ask: Is there a relationship between a country's general PISA score and its IMO performance? How do a country's mathematics, science, and reading scores on the PISA correlate with Olympiad success? Additionally, we explore the role of gender representation on national IMO teams by analyzing whether team gender composition influences outcomes. Finally, we will put it all together by trying to answer our main question: What factors impact a teams performance most at the IMO? Through this analysis, we aim to uncover how broader educational quality, national priorities, and team dynamics intersect in shaping international achievement in mathematics.

The Data Sets

The International Mathematical Olympiad (IMO) data we will be using is from Tidy Tuesday, which gives us every competition ever held starting in 1959 and going up to this year's 2024 competition (Khurana, 2024). This data set includes 18 different variables/columns, with over 3700 values/observations. The data is organized by year and also by each country that participated in that year's competition. It includes integer variables and values that give us the names of each country and the names of the leaders of each team. It also gives us numeric values that give us the number of competitors on each team, how many of each gender they are, how many points they got on each selected problem, and how many medals and what kind each country received in that year's competition.

We got our PISA data from their official website, which provided an overview of the results from a test conducted by PISA (OECD). There are 6 different variables/columns, with a total of 80 observations/values. They are all numeric values that give us an overall score for their PISA rating, along with individual scores for each subject, including math, science, and reading. Because this is not the main data set we will be using to work with, we will combine it with our other data set by Country since both contain the same countries that have participated in the IMO. In our analysis, we use the average national scores as listed in the "PISA MATH 2022" column as a quantitative indicator of mathematical proficiency, which we will then compare to each country's performance during the International Mathematical Olympiad.

Results

Relationship Between International Math Competition Performance and PISA Scores

Analysis of our Three Countries of Focus Our Data and Results

- **PISA 2022 Results:** International standardized test scores for students (Math, Science, Reading, Overall scores).
- **IMO 2022 Results:** Scores from the International Mathematical Olympiad.

We began by gathering and preparing two datasets: one containing national PISA scores, and another capturing the results of each country's IMO team. To ensure comparability, we standardized the country names across both datasets and filtered the data to focus solely on results from the year 2022.

flagCode	OverallPisaScore2022	PISAScoresMathScore2022	PISAScoresScienceScore2022	PISAScoresReadingScore2022	Team_IMO_Score
CN	1605	552	543	510	252
US	1468	465	499	504	207
NL	1440	493	488	459	145

From the IMO data, we calculated a single "Team IMO Score" for each country by summing the points earned across all competition problems. For the PISA data, we used the scores already provided in the four domains: overall performance, mathematics, science, and reading. After preparing the datasets, we merged them into a single combined table, linking each country's IMO performance with its corresponding PISA results. We first computed the correlation between each country's IMO team score and its PISA scores across all four domains.

Correlation	Value
Overall Pisa 2022 vs IMO	0.8979
Math Pisa 2022 vs IMO	0.5932

Science Pisa 2022 vs IMO	0.9111
Reading Pisa 2022 vs IMO	0.9479

Interpretation of Results

The strongest relationship is observed with the PISA Reading Score 2022 (0.948), suggesting (emphasis on ‘suggesting’, since we are only basing this on an analysis of three countries at the moment) that countries with higher reading proficiency among students tend to also perform better at the International Mathematical Olympiad.

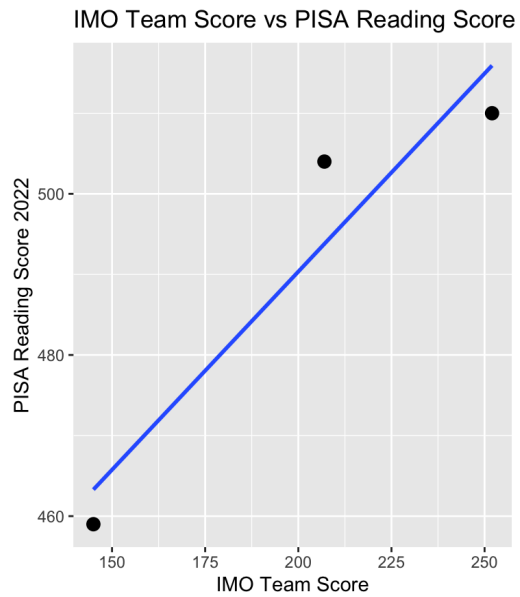
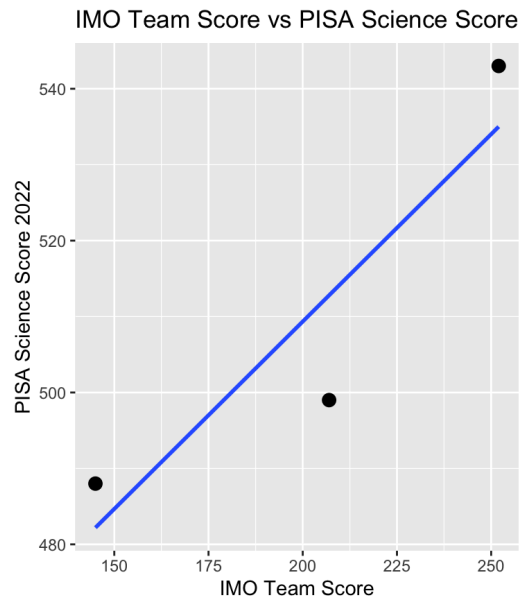
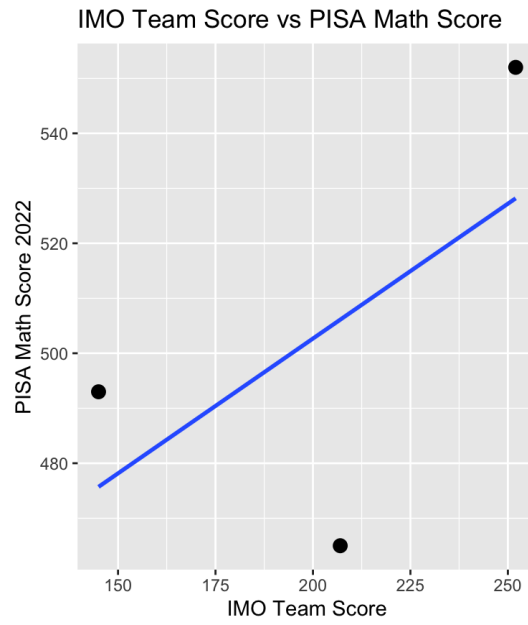
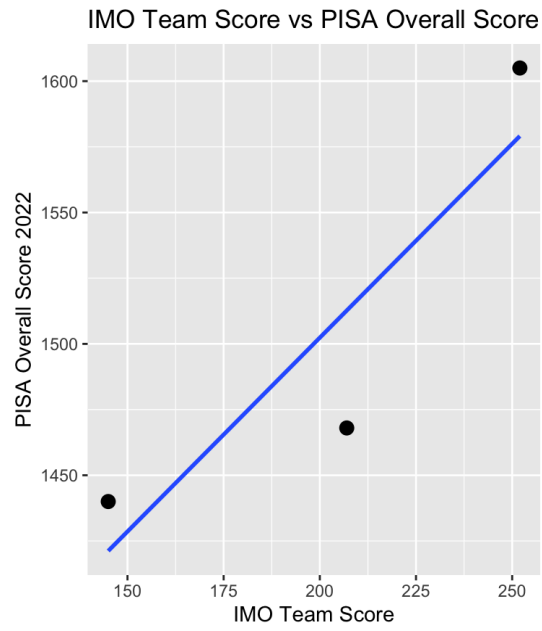
A similarly strong correlation exists with the PISA Science Score 2022 (0.911), indicating that strong science skills are also closely associated with mathematical problem-solving ability.

The correlation between the Team IMO Score and the Overall PISA Score 2022 is slightly lower (0.898) but still strong. This suggests that while general academic proficiency is related to IMO success, specific subject strengths matter more.

Interestingly, the PISA Math Score 2022 shows a moderate correlation (0.593) with the Team IMO Score, weaker than expected given the mathematical nature of the IMO. This may indicate that the IMO measures more advanced or specialized mathematical abilities than what is captured by standard PISA math assessments.

One possible explanation lies in the PISA and IMO participants: PISA tests 15-year-old students from the general school population. IMO contestants are typically older (16–19 years) and represent the absolute top of mathematical talent in each country, often selected through multiple national competitions.

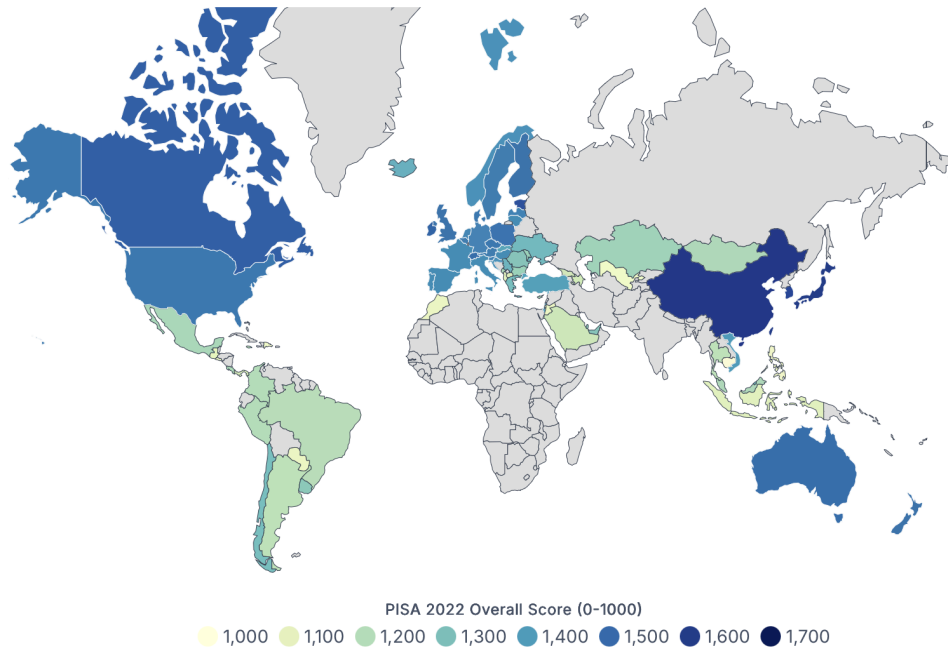
Graphs



Global Analysis of PISA Scores

Our Data and Results

2022



Initially, our analysis focused on a subset of three countries—the United States, the Netherlands, and China—but we later expanded the scope to include all countries with available data for both PISA and IMO. We calculated correlation coefficients between IMO scores for all countries for the years 2022 and 2018, in addition to the analysis of our restricted dataset. Our 2022 correlation result is based on 70 countries, as these were the countries included in both the 2022 PISA assessment and the 2022 IMO competition.

Correlation	Value All Countries 2022	Value 3 Countries (USA, NL, CN) 2022
Overall PISA Score vs IMO	0.3975	0.8979
Math PISA Score vs IMO	0.4283	0.5932
Science PISA Score vs IMO	0.3916	0.9111
Reading PISA Score vs IMO	0.3555	0.9479

2018

The PISA 2018 study had a more limited scope. In comparison to our 2022 data, fewer IMO-participating countries were part of the PISA assessment in 2018. As a result, the available data for 2018 includes fewer observations. So, our correlation result is based on the following 23 countries, as these were the countries included in both the 2018 PISA assessment and the 2018 IMO competition:

 People's Republic of China	 Singapore	 Estonia
 Japan	 Canada	 Finland
 Poland	 Ireland	 United Kingdom
 Slovenia	 New Zealand	 Netherlands!
 Denmark	 Germany	 Belgium
 Australia	 Switzerland	 Norway
 Czech Republic	 United States	 France
 Portugal	 Austria	

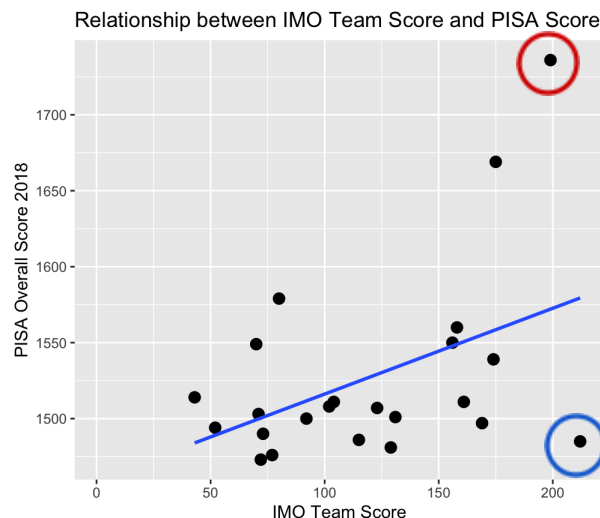
In the 2018-analysis we focus only on the **overall PISA score**, as domain-specific scores (math, science, and reading) were not readily available in our original dataset/source and were furthermore not easy to extract from other sources, because they only showed the scoretables as an image instead of actual data in text-formatting.

Correlation	Value All Countries 2022	Value All Countries 2018
Overall PISA Score vs IMO	0.3975	0.4393

Interpretation of Results

In the three-country subset, the correlations between IMO team scores and PISA results for the year 2022 were extremely strong (above 0.89 in most domains), indicating that in these specific high-performing countries, exceptional performance during the IMO is closely aligned with strong overall PISA scores. However, when the analysis is broadened to include all participating countries, the correlations decline significantly, suggesting that the relationship between IMO and PISA performance is not universal. In many countries, success in competitions like the IMO may depend on training for a small group of students, rather than reflecting the educational (PISA) level of the broader student population. Meanwhile, countries like the USA, Netherlands, and China not only invest in their top performers but also maintain high overall standards across subjects (as they have the means to do so), which leads to stronger alignment between IMO and PISA outcomes.

In 2018, the correlation between IMO team scores and overall PISA scores was 0.4393, which indicates a moderate positive relationship. In comparison, the 2022 correlation was slightly lower at 0.3975, suggesting a weaker but still positive association. This difference is probably due to the relatively smaller sample size (23) regarding our 2018 data. In our 2018 correlation analysis, China and the USA clearly emerged as statistical outliers among the 23 data points, which happen to be 2 of the 3 countries we had analyzed previously. Quite interesting.



 **China (2018):**

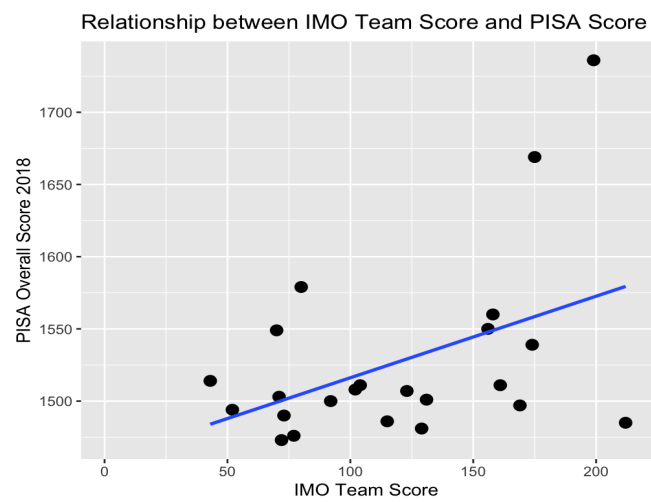
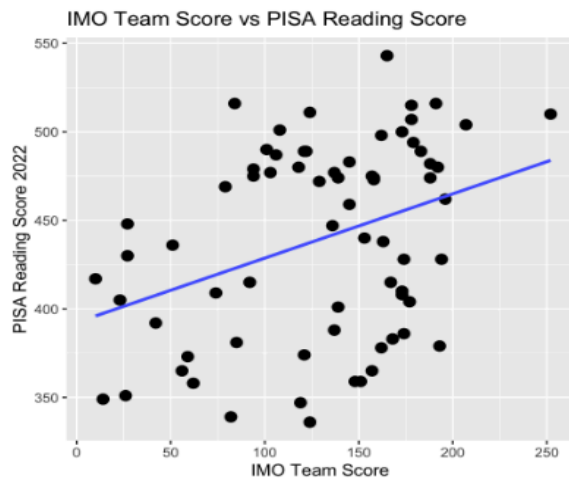
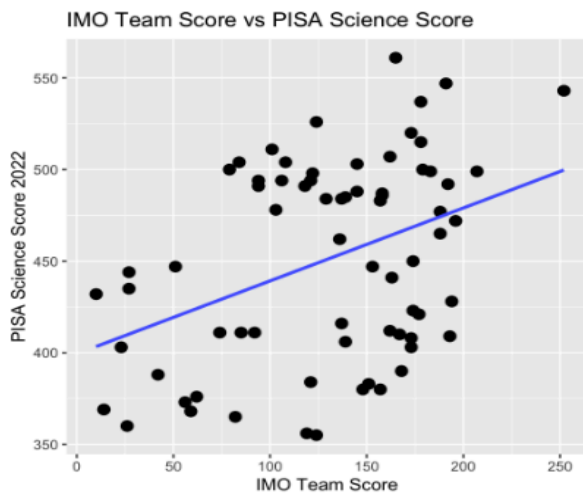
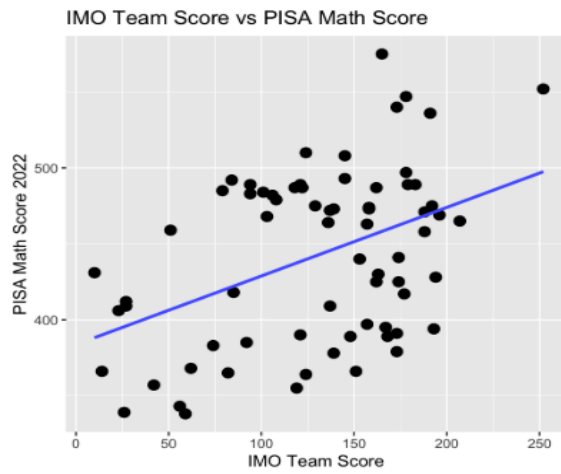
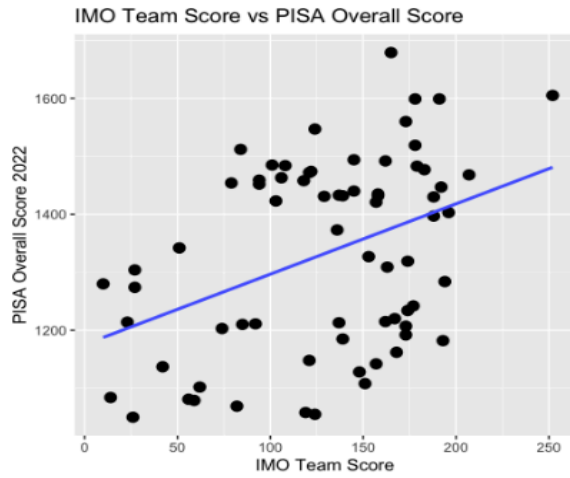
PISA Overall Score: 1736
IMO Team Score: 199

 **United States (2018):**

PISA Overall Score: 1485
IMO Team Score: 212

As visualized in the scatterplot above, China and the United States emerge as clear outliers in the relationship between IMO team scores and PISA overall scores in 2018. China, circled in the top-right corner, has both a high PISA and IMO score, extremely high, even. In contrast, the United States, circled in the lower-right, has the highest IMO score, making their team the winner of the 2018 IMO competition, despite only average overall PISA performance.

Graphs



Analyzing Gender Diversity and Outcomes at IMO

Introduction for Sample Countries

In this section of our report, we are switching to focus on the IMO data and see how gender diversity is reported and the outcomes because of it. We aim to see if there is a difference in Team IMO Score when there is a female contestant on the team compared to when there is no female contestant. In an article on gender differences and scholastic achievements, co-author Susan Voyer says, “The fact that females generally perform better than their male counterparts throughout what is essentially mandatory schooling in most countries seems to be a well-kept secret, considering how little attention it has received as a global phenomenon”. We see this as a great baseline to work with and a good reason to have a preliminary expectation that when there are female contestants on the team, the average Team IMO Score would be greater than one with all males. We will be testing this theory out by creating and using hypothesis tests in our analysis.

Data

To begin, after looking over the data, we decided to mutate the problem score variables and sum them up to create the main variable that we will be working with: “Team_IMO_Score”. We then also decided to take all the total teams and instead of having each total female count on each team, we mutated each team to either have a “Female Present” or “No Female” so it would be easier for us to work with. For our initial analysis, we decided to create a sample dataset that would only look at 3 countries of our choice: the United States of America, China, and the Netherlands.

Results and Interpretations

Our initial thought was to first get an idea of what we were working with in terms of total teams with and without females, so we created a table that included each country and their team amounts.

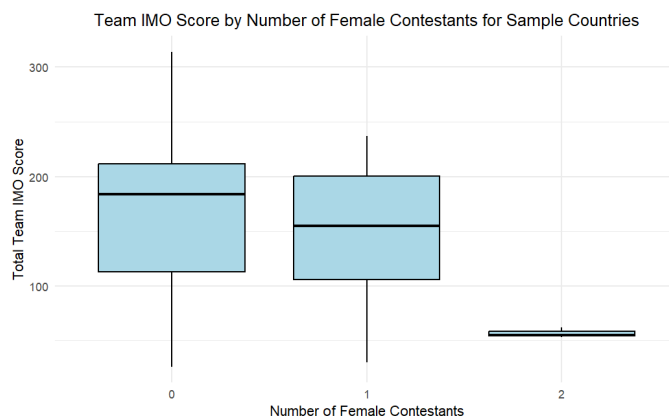
Country	Count of Teams All Time With 0 Female Contestants	Count of Teams All Time With At Least 1 Female Contestant
China	30	9
USA	42	6
Netherlands	38	15

As we can see from the table, it is clear that there is some bias in all these countries that involves not having many females on competing teams. In seeing this, we then decided to create our null hypothesis - H_0 - *There is no difference in means in Team IMO Score when there are female contestants representing*, and also create our alternative hypothesis - H_1 - *There is a difference in means in Team IMO Score when there are female contestants representing*. We used these to correctly interpret the results of our t-test for each country, trying to determine the answer to our question.

T-Test and Box Plot for Our Sample Countries

Country	Mean (Females Present)	Mean (No Females)	Confidence Interval	P - value
China	204.44	205.53	(-24.27, 22.09)	.923
USA	184.67	195.29	(-39.20, 17.97)	.410
Netherlands	90.60	95.32	(-27.73, 18.30)	.679

Interpreting our results, we can see that each hypothesis test gave us very similar means for each country, a confidence interval that went through 0, and a non statistically significant p-value, this leads us to fail to reject the null hypothesis that said that there was no difference in means when there was a female competitor compared to no female competitors. We then created a box plot to look at the scores visually to see how they compare to each other.



We can see that for the teams that have no females compared to teams that have at least one female, their means are very similar for our sample data of countries.

Our results tell us that our preliminary conclusion was actually incorrect, and that having at least one female on the team does not improve or hurt the team's Total IMO Score. This was shocking for us but not totally unexpected. We saw from the table before that there were barely any teams all time that had included at least one female competitor, this leads us to know that there was major bias within the total number of teams competing. We also know that our sample

size for 3 countries is super small, and we can't make any official conclusions because of it. So our next goal is to look at all the countries on a broader scale.

Introduction for Worldwide Countries

In this section of our analysis of gender diversity, we will switch from our sample size of 3 countries, to all countries we have - 139 countries - to try and get a better understanding if our preliminary expectation was wrong and if our question could be answered.

Results and Interpretations

To do this, we once again started by creating a table to see if the past results for our sample data would still hold true for the amount of teams with and without females.

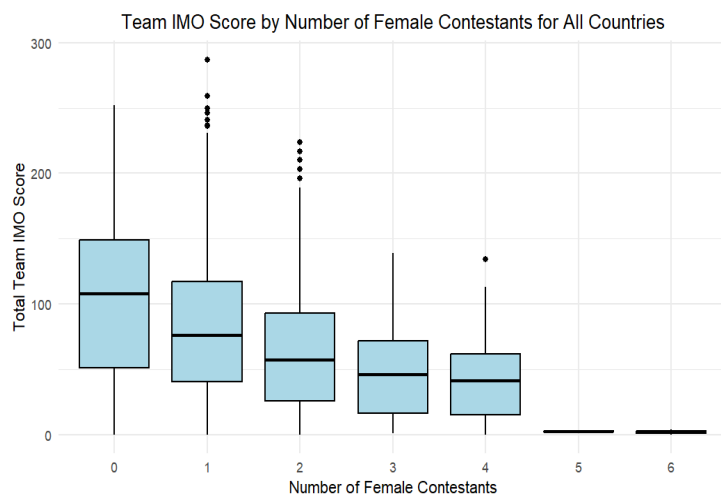
Count of Females All Time With 0 Female Contestants	Count of Females All Time With At Least One Female Contestant
2484	1296

The table for all countries gives us the same results as our sample countries did. There are basically 2 times more teams all time with 0 females compared to teams with at least 1 female. This is giving us more proof to see that there has been tons of bias all time between gender diversity. Once again, we then decided to create our null hypothesis - *H₀ - There is no difference in means in Team IMO Score when there are female contestants representing*, and also create our alternative hypothesis - *H₁ - There is a difference in means in Team IMO Score when there are female contestants representing*. We used these to correctly interpret the results of our t-test now for all countries combined to try and see if our expectation is correct or not.

T-Test and Box Plot for All Countries

Country	Mean (Females Present)	Mean (No Females)	Confidence Interval	P - value
ALL	77.19	102.87	(-32.81, -18.55)	p < .001

Interpreting our results, not only was our preliminary expectation wrong, but it was the complete opposite of what we thought would show. With the two means being very different, a confidence interval that does not go through 0, and a very small, statistically significant p - value, we can conclude that we need to reject the null hypothesis that says that there would be no difference in Team IMO Score when there was at least one female present on the team. We then created a box plot to visually analyze this result to see if there were any outstanding factors.



We can tell that all time, the more female contestants that you add to a team, seems to bring down the average Total Team IMO Score which is contradictory to our expected beliefs. While teams with 1 female have spikes and outliers that are greater than the teams with 0 females, the teams with 0 females just seemed to score better on average than any team including at least one female.

We can conclude from our testing that our preliminary expectation was not only incorrect, but completely opposite then what we thought. It seems very odd that the scores on average are not only different, but there is a pretty big difference between the mean Team IMO Scores when there are no females compared to when there are at least one. Even with the potential bias of there being almost 2 times as many no female teams compared to at least 1 female teams, the data shows that the more females you add to the team, the lower score you seem to get in the competition. The results are just too clear and too statistically significant for us to believe otherwise.

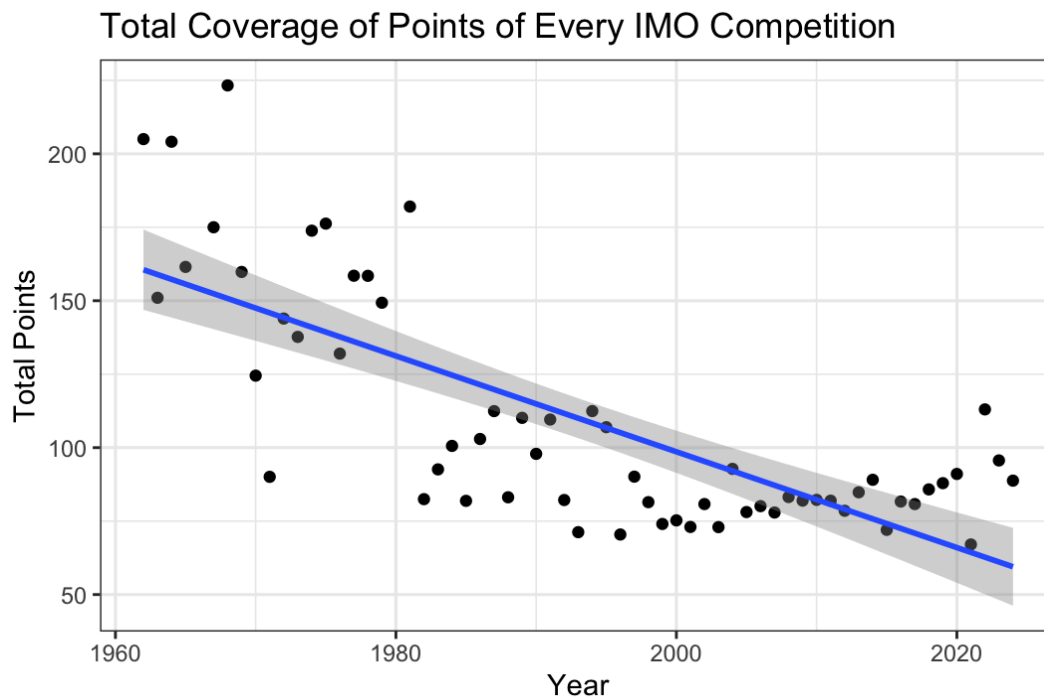
Predicting Scores in Future Competitions

Introduction for Full Analysis of Scores

For this portion of the analysis, we wish to understand what the future of the International Mathematical Olympiad will bring. To begin this, we are using the previous total score tallies for the IMO competition, which is composed of all the points from the six questions added together. With the total scores, we will analyze how the average score fluctuates over time. Because there are decades of competition, this will provide enough information to create a linear regression for a maximum likelihood estimation. This estimation will create approximate values for future competition years based on the history of the data.

Graphs and Regression

For the first part of the analysis, we created a plot that shows the total points averaged from each IMO competition over the decades. It was built by creating a new dataset, appropriately named `mean_per_year`, that grouped the data based on the year and took the mean of the points from those years. The resulting dataset had 61 observations for the years with data available for the scores. This graph is going to be the basis of the regression analysis as it will be able to visualize the trend within the points.



This graph shows the total coverage of points from each IMO competition. The x-axis represents the years the competition took place, and the y-axis shows the average points during that year. The blue line is the line of best fit to show the trend of how the total number of points changes over time. The years 1959, 1960, 1961, 1966, and 1980 were removed due to a lack of score data.

This graph shows a visibly decreasing trend in the points over time, suggesting that a future analysis of the data may show a continued decrease in the average scores of the competition. To understand this graph statistically, we ran a regression model of the total points against the years:

Variable	Result	P-value
Intercept	3359.16	$p < 0.001$
Slope	-1.6303	$p < 0.001$

This table shows the two variables and their estimated resulting value with the corresponding p-value. Both of these values received p-values that are statistically significant in our data. It is also supported with an adjusted R^2 value of 0.5553 showing over half of the data can be described with this model. This is significant for this prediction since it is necessary for a majority of the data to be incorporated with a model, even more so when it is given between a single variable model giving stronger importance to that predictor. Given the number of years, this information gives us plausible values to work with for a prediction, as there is enough evidence to support values based on this regression model. It is important to see significant evidence if we are to believe our predictions to be reasonable, given the data from the past competitions.

Prediction Data and Analysis

The next step of the analysis is to use our information to create a predictive function. This function will be built as a slope-intercept function because the regression created was a linear model of the data. To start, the dataset mean_per_year was taken, and two variables named x and y, of the year and mean scores, respectively. The key to creating a prediction function off of a linear model is using a negative-log-likelihood function. This is because we are trying to optimize the scores in our dataset, which automatically minimizes whatever function it is given. To combat this, creating a negative likelihood function will also minimize our data to provide a maximized outcome for our likelihood estimation. It will in the end create a function looking like:

$$y = \beta_0 + \beta_1 * x$$

Where y is the predicted score and x is the years. The beta variables are fillers for the intercept and slope of the function.

To properly run the optimization function, we also needed to come up with starting terms that will drive the equation to create the predictions. These will be pulled from the linear

regression model and input into the slope and intercept portions of the slope-intercept form equation.

Finally, when all is put together in the function, the fit parameters can be used to create a prediction model. When all comes together, the equation comes out to:

$$Points = 3359.16 - 1.63 * year$$

And using it over the next five years creates the following results:

Year	Score
2025	57.834
2026	56.204
2027	54.573
2028	52.943
2029	51.313

When looking over this information, the original graph showed a decreasing trend line in the data. Then, with a regression model fit to it, it showed statistically significant evidence of this model fitting the scores over the years. Finally, the next five years of predictions continue to show a decreasing trend which is supported by the testing previously. This creates an interesting prediction where the overall average of each competition will begin to lower as time continues on. Looking at the graph as a whole, this may or may not be the truth since the scores have definitely changed since the beginning of the competition, and it may seem as if the scores have plateaued after a certain point. One way this could happen may be a change in the types of questions over time as the competition has grown from the beginning. If the questions have developed to be more focused and each competition can be more prepared for, there may be the causing factor in if the points are truly going decreasing or remaining more constant in recent years. Overall, with the models created, the prediction shows a decrease in the average scores per year for the next five years.

What Impacts A Country's Performance at the IMO?

Introduction for Factors that Impact Performance

For the final part of the project, we aim to put all of our data together in hopes of figuring out if these factors that we analyzed have a true impact on a team's performance at the IMO. We do this by creating a multiple linear regression model in order to see what effects the variables have on a team's score.

Regression Model 1 and Interpretation

Our first thought was to just put all of the variables that we used into a model and see what effects it brought. We used the variables: team_size_female, team_size_male, OverallPisaScore2022, and PISAScoresOverallScore2018 in our first model.

Variables	Estimate	P - value
(Intercept) - Team_IMO_Score	-269.594	.002
team_size_female	6.477	.241
team_size_male	29.409	p < .001
OverallPisaScore2022	.233	.004
PisaScoresOverallScore2018	-0.090	.386

This model gave us an *Adjusted R-squared* of .2135, which is not great, but is a good starting point to have about 21% of the variance in the Team_IMO_Score explained by these variables. We can see that for the Team-IMO_Score, when all predictors are 0, the expected outcome is -269.594, which is not possible, but is also statistically significant with a p - value of .002. For team_size_female, for every female that is added to the competition team, the predicted Team_IMO_Score only increases an estimated 6.477, which shows that the addition of a female does not really make too much of a difference and is not statistically significant with a p - value of .241. For team_size_male, for every male that is added to the competition team, the predicted Team_IMO_Score increases an estimated 29.409, which is good and also statistically significant with a p - value of less than .001. For OverallPisaScore2022, for every 1-point increase in a country's 2022 PISA score, the predicted Team_IMO_Score increases by 0.233 points, which is also statistically significant with a p - value of .004. Finally, for PisaScoresOverallScore2018, a 1-point increase in 2018 PISA score, the predicted Team_IMO_Score would decrease by .090, this shows that this variable does not play a significant role and is not statistically significant

with a p - value of .386. Although this model is not great, we believe that with transformations we can increase the Adjusted R-squared score even more.

Final Regression Model and Interpretation

After multiple attempts to maximize our Adjusted R-squared value, we were able to increase and maximize our value by using transformations and adding new variables to the model. We used the variables: team_size_male, team_size_female, and PISAScoresOverallScore2018 normally without any transformations. We used these old and new variables in a transformed way by taking the square root: OverallPisaScore2022, PISAScoresMathScore2022, PISAScoresReadingScore2022, and PISAScoresScienceScore2022.

Variables	Estimate	P - value
(Intercept) - Team_IMO_Score	-793.417	p < .001
team_size_male	29.176	p < .001
team_size_female	8.550	.111
OverallPisaScore2022 (sqrt)	-9666.580	p < .001
PISAScoresOverallScore2018	-0.175	.111
PISAScoresMathScore2022 (sqrt)	5586.526	p < .001
PISAScoresReadingScore2022 (sqrt)	5543.980	p < .001
PISAScoresScienceScore2022 (sqrt)	5657.315	p < .001

This model gave us a new and maximized *Adjusted R-squared* of .2656, which again is not great, but any improvement in real life data is good, so we are happy with our findings. Even though only 26% of the variance in the Team_IMO_Score is explained by these variables, we know that with the data and variables we had, we could only do so much in predicting a competing teams performance IMO score. We can see that for the Team-IMO_Score, when all predictors are 0, the expected outcome is -793.417, which is also not possible, but is also statistically significant with a p - value of less than .001. For team_size_male, for every additional male that is added to the team, the predicted Team_IMO_Score is increased by

29.176, and is also statistically significant with a p - value of less than .001. For team_size_female, each additional female increases the predicted Team_IMO_Score by 8.550, but we cannot say that this variable has any real effect since it is not statistically significant with a p - value of .111. For OverallPisaScore2022 (sqrt), there is a very strong negative correlation with the Team_IMO_Score because for every overall point added, the Team_IMO_Score decreases by -9666.580. Although this number is large because of the square root transformation, the effect on the prediction is statistically significant with a p - value of less than .001. For PISAScoresOverallScore2018, each 1-point increase in the 2018 PISA score decreases the prediction for the Team_IMO_Score by -0.175, but this is not statistically significant because of the p - value of .111. For PISAScoresMathScore2022 (sqrt), there is a very strong positive correlation with the Team_IMO_Score because for every overall point added, the Team_IMO_Score increases by 5586.526. Although this number is large because of the square root transformation, the effect on the prediction is statistically significant with a p - value of less than .001. For PISAScoresReadingScore2022 (sqrt), there is a very strong positive correlation with the Team_IMO_Score because for every overall point added, the Team_IMO_Score increases by 5543.980. Although this number is large because of the square root transformation, the effect on the prediction is statistically significant with a p - value of less than .001. For PISAScoresScienceScore2022 (sqrt), there is a very strong positive correlation with the Team_IMO_Score because for every overall point added, the Team_IMO_Score increases by 5657.315. Although this number is large because of the square root transformation, the effect on the prediction is statistically significant with a p - value of less than .001. Although this model does not have a great Adjusted R-squared value, we have used transformations to increase our score and have done the best we could with the given variables to try and explain the variance of the Team_IMO_Score variable.

Conclusions

In this project, we investigated the relationships between general academic performance within a country, gender representation, and performance at the International Mathematical Olympiad. First of all, we linked PISA scores with IMO performance for both a focused group of countries and a broader global dataset. We found that there is quite a strong correlation between these scores when analyzing China, the United States and the Netherlands specifically, but that this relationship weakens significantly when including all the countries that participated in both the IMO and PISA assessment in the years 2018 and 2022. This suggests that IMO performance is most likely driven by specialized training for participants rather than the actual education system.

Our analysis of gender diversity in IMO teams revealed surprising results. Contrary to our preliminary expectations, teams with female participants tended to have lower average scores than all-male teams, and this was specifically noticeable when analyzing the Netherlands. Even though our findings were quite clear regarding this matter, they should be interpreted with caution due to systemic gender imbalances in participation and the historical underrepresentation of female competitors.

Furthermore, our trend analysis of IMO performance over time indicated a gradual decline in average team scores. Our regression model predicted that this trend may continue in the coming years and could be due to changing problem difficulty.

Overall, while our models explained only a modest portion of the variance in IMO performance, they did provide us with meaningful insights and definitely challenged us to think about how and why our results do not align as smoothly with our preliminary results as we hoped they would.

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